

Complex Numbers

Basics

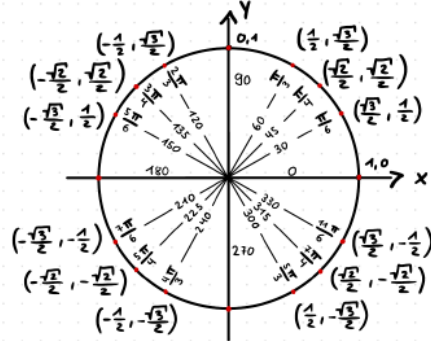
- Def :** $z = a + bi \Leftrightarrow \Re(z) = a, \Im(z) = b$
Def : $z = a + bi \Leftrightarrow \bar{z} = a - bi \Leftrightarrow r \cdot e^{2\pi - \phi}$
Def : $z = r \cdot \cos(\phi) + i \cdot \sin(\phi)$
Def : $|z| = r = \sqrt{x^2 + y^2} = \sqrt{z \cdot \bar{z}}$
Def : $\phi = \begin{cases} \arctan \frac{y}{x} & 1. \text{ Q} \\ \arctan \frac{y}{x} + \pi & 2./3. \text{ Q} \\ \arctan \frac{y}{x} + 2\pi & 4. \text{ Q} \end{cases}$

Operations

- Def :** $z_1 \pm z_2 : (x_1 + x_2) \pm i(y_1 + y_2)$
Def : $z_1 \cdot z_2 : (x_1 + i \cdot y_1) + (x_2 + i \cdot y_2) = r_1 \cdot r_2 e^{i(\phi_1 + \phi_2)}$
Def : $\frac{z_1}{z_2} : \frac{r_1}{r_2} e^{i(\phi_1 - \phi_2)} = \frac{z_1 \cdot \bar{z}_2}{|z_2|^2}$
Def : $\sqrt[n]{a} \Leftrightarrow a = z^n \Leftrightarrow |a| \cdot e^{i\phi} = r^n \cdot e^{i\omega n} \Leftrightarrow r = \sqrt[n]{|a|}, \omega = \frac{\phi + 2k\pi}{n}$

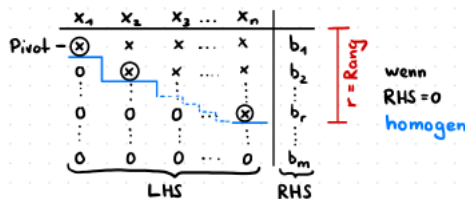
Polynomials

The roots of a complex polynomial are pairwise conjugated. **Def :** $z = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$
Def : $az^n + c = 0 \Leftrightarrow z = \sqrt[n]{-\frac{c}{a}}$



SLE

Gauss Algorithm



- Compability Conditions:** $b_{r+1} = \dots = b_m = 0$
S 1.1: $Ax = b$ hat min eine Lösung $\Leftrightarrow r = m$ oder $r < m + VB$ dann: $r = n \Leftrightarrow 1$ Lösung, $r < n \Leftrightarrow \infty$ Lösungen
Cor 1.7: For a quadratic SLE with n equations and n variables we have the following set of equivalence, of which ONLY one of them can be true; So, EITHER
 i Rank(A) = n (A is regular)
 ii for every b there exist at least one solution
 iii for every b there exists exactly one solution
 iv the corresponding homogeneous system has only the trivial solution
 OR the following equivalences hold
 v Rank(A) < n (A is singular)
 vi for some b there exists no solution
 vii for no b a unique solution exists
 viii for some b infinity many solution exists
 ix the corresponding homogeneous system has non-trivial solutions

Matrices and Vectors

Definitions

- A $m \times n$ matrix hat m row (Zeilen)↓ and n columns (Spalten)→, in which the i,j element gets noted by $a_{i,j}$ or $(A)_{i,j}$
Def nullmatrix: Has in every entry 0
Def diagonalmatrix: Has in every entry 0 except for the diagonal: $(D)_{ij} = 0$ for $i \neq j$ one can write $\text{Diag}(d_{11}, \dots, d_{nn})$
Def identity: The identity is written as $I_n = \text{Diag}(1, \dots, 1)$ It holds that $AI = IA = A$
Def upper triangular matrix: We have $(R)_{ij} = 0$ for $i > j$ (Rechtsdreiecksmatrix)
Def lower triangular matrix: We have $(L)_{ij} = 0$ for $i < j$ (Linksdreiecksmatrix)
Def Matrix-set: The set of $m \times n$ -matrices is written as: $\mathbb{E}^{m \times n}$ For vectors we have: $\mathbb{E}^n$, where $\mathbb{E}$ is $\mathbb{R}$ or $\mathbb{C}$
Def matrix multiplication: If $C = AB$ then one can write
 $C_{ij} = (AB)_{ij} = \sum_{k=1}^n (A)_{ik} (B)_{kj} = \sum_{k=1}^n a_{ik} b_{kj}$
S 2.1:
 - $(\alpha\beta)A = \alpha(\beta A)$
 - $(A+B) \cdot C = A \cdot (B+C)$
 - $(\alpha A)B = \alpha(AB)$
 - $(AB) \cdot C = A \cdot (BC)$
 - $(\alpha + \beta)A = \alpha A + \beta A$
 - $(A+B) \cdot C = AC + BC$
 - $A \cdot (B+C) = AB + AC$
 - $\alpha(A+B) = \alpha A + \alpha B$
 - $A+B = B+A$**S 2.20:** Let A and B be unitary(orthogonal). It holds:

- A is regular and $A^{-1} = A^H (A^T)$
 - $AA^H (AA^T) = I$
 - A^{-1} is unitary (orthogonal)
 - AB is unitary (orthogonal)
- Def Zerodiviser:** If $AB = 0 \Leftrightarrow A, B$
Zerodiviser, Nullteiler
Def transposes: $(A^T)_{ij} = A_{ji}$
Def conjugate transposed: $A^H = (\bar{A})^T = \overline{A^T}$
Def symmetric: $A^T = A \Leftrightarrow A$ symmetric
Def skew-symmetric: $A^T = -A \Leftrightarrow A$ skew-symmetric
Def hermitian: $A^H = A \Leftrightarrow A$ hermitian
S 2.6: Also accounts for A^T instead of A^H . $\bar{\alpha}$ simplifies to α
 - $(A^H)^H = A$
 - $(\alpha A)^H = \bar{\alpha} A^H$
 - $(A+B)^H = A^H + B^H$
 - $(AB)^H = B^H A^H$**S 2.7:** For symmetric matrices A and B it holds that: $AB = BA \Leftrightarrow AB$ is symmetric It holds for arbitrary matrix C that: $C^T C$ and CC^T are symmetric. The same holds for the hermitian case

Scalarproduct and Norm

- Def euclidian scalarproduct:**
 $\langle x, y \rangle = x^T y = \sum_{k=1}^n \bar{x}_k \cdot y_k \xrightarrow{\mathbb{R}} \sum_{k=1}^n x_k \cdot y_k$
S 2.9:
 - S1** $\langle x, y+z \rangle = \langle x, y \rangle + \langle x, z \rangle$ (linear in 2nd factor)
 - S1** $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$ (linear in 2nd factor)
 - S2** for $\mathbb{E} = \mathbb{R}$:
 $\langle x, y \rangle = \langle y, x \rangle$ (symmetric)
 - S2'** for $\mathbb{E} = \mathbb{C}$:
 $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (hermitian)
 - S3** $\langle x, x \rangle > 0, \langle x, x \rangle = 0 \Leftrightarrow x = 0$ (positiv definite)**Cor 2.10:**
 - S4** for $\mathbb{E} = \mathbb{R}$: linear in 1st factor
 $\langle w+x, y \rangle = \langle w, y \rangle + \langle x, y \rangle$
 $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
 - S4'** for $\mathbb{E} = \mathbb{C}$: conjugate-linear in 1st factor
 $\langle w+x, y \rangle = \langle w, y \rangle + \langle x, y \rangle$
 $\langle \alpha x, y \rangle = \bar{\alpha} \langle x, y \rangle$**Def norm:** $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^T x} = \sqrt{\sum_{k=1}^n (|x_k|)^2} \xrightarrow{\mathbb{R}} \sqrt{\sum_{k=1}^n x_k^2}$

- S 2.11:** $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$ (Cauchy-Schwarz inequality, " " holds when y is a multiple of x or vice versa)
Def CBS: CBS is a property of the scalar product: CBS squared yields:
 $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$
S 2.12: For the euclidian norm holds:
 - N1** $\|x\| > 0, \|x\| = 0 \Leftrightarrow x = 0$ (positiv definit)
 - N2** $\|\alpha x\| = |\alpha| \|x\|$ (homogeneous)
 - N3** $\|x \pm y\| \leq \|x\| + \|y\|$ (Triangle-inequality)**Def :** Angle ϕ between x, y :
 $\phi = \arccos \frac{\text{Re}(\langle x, y \rangle)}{\|x\| \cdot \|y\|} \xrightarrow{\mathbb{R}} \arccos \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}$
Def : x, y are orthogonal: $\langle x, y \rangle = 0 \Leftrightarrow x \perp y$
S 2.13: $\|x \pm y\|^2 = \|x\|^2 + \|y\|^2 \Leftrightarrow x \perp y$ (Pythagoras) **Def p-norm:**
 $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}}$

Outer Product and Projections

- Def outer product:** m-vector x and n-vector y : xy^T
S 2.14: A $m \times n$ -matrix has rank 1 if it is the outer product of an m-vector $\neq 0$ and n-vector $\neq 0$
S 2.15: The orthogonal projection $P_y x$ of the n-vector x onto y is defined as:
 $P_y x = \frac{1}{\|y\|^2} \cdot y y^H x = u u^H = P_u$ where $u = \frac{y}{\|y\|}$
Def projections matrix: $P_y = \frac{1}{\|y\|^2} \cdot y y^H$ It has the properties: $P_y^H = P_y$ (hermitian/symmetric) and $P_y^2 = P_y$ (idempotent)

Inverse

- Def invertible:** $\exists A^{-1} \Leftrightarrow A^{-1} \cdot A = A \cdot A^{-1} = I$
S 2.17: A is invertible $\Leftrightarrow \exists X : AX = I \Leftrightarrow X$ is unique $\Leftrightarrow A$ is regular
S 2.18: If A, B are regular:
 - A^{-1} is regular and $A^{-1-1} = A$
 - AB is regular and $(AB)^{-1} = B^{-1} A^{-1}$
 - A^H is regular and $(A^H)^{-1} = (A^{-1})^H$**S 2.19:** If A is regular das LGS $Ax = b$ has the unique solution $x = A^{-1} b$
 Finding an inverse $[A|I] \xrightarrow[\text{op}]{\text{row}} [I|A^{-1}]$ if

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ and } \det(A) \neq 0 \Leftrightarrow A \text{ is invertible}$$

$$\Leftrightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$A = \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right] \Leftrightarrow A^{-1} \left[\begin{array}{cc|cc} a_{11}^{-1} & a_{12}^{-1} & 1 & 0 \\ a_{21}^{-1} & a_{22}^{-1} & 0 & 1 \end{array} \right]$$

Orthogonal and unitary matrices

- Def unitary/orthogonal:** $AA^H = I, AA^T = I \Leftrightarrow A$ is unitary/orthogonal $\Leftrightarrow \det(A) = \pm 1$
S 2.20: A, B are unitary/orthonormal:
 - A is regular and $A^{-1} = A^H$
 - $AA^H = I_n$
 - A^{-1} is unitary/orthogonal
 - AB is unitary/orthogonal
 - columns are orthonormal**S 2.21:** Images from unitary/orthonormal matrices are conformal (längen-winkeltreu)

Def 2d rotation: $R(\phi) = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix}$

Def 3d rotation:
 $R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}, R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}, R_z(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$

LU-Decomposition

- The LU-decomposition is useful when multiple SLE have the same A
- Find $PA = LR$
 - solve $Lc = Pb$
 - solve $Rx = c$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -4 \\ 0 & 1 & 0 & -4 & -8 & 43 \\ 0 & 0 & 1 & 2 & -5 & -3 \end{array} \middle| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \xrightarrow{+4I} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -4 \\ 0 & 1 & 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 0 & -3 & -4 \end{array} \middle| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \xrightarrow{-2I} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -4 \\ 0 & 1 & 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 0 & -3 & -4 \end{array} \middle| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \xrightarrow{III} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -4 \\ 0 & 1 & 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 0 & -3 & -4 \end{array} \middle| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \xrightarrow{I} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -4 \\ 0 & 1 & 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 0 & -3 & -4 \end{array} \middle| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)$$

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III

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P U (R) L (U)

Vectorspaces

Def : A vectorspace V over $\mathbb{K}$ is a non-empty set, on which vectoraddition and scalarmultiplication is defined

Def Axioms:

- V1 : $x + y = y + x$
- V2 : $(x + y) + z = x + (y + z)$
- V3 : $\exists 0 \in V : x + 0 = x$
- V4 : $\forall x \exists -x : x + (-x) = 0$
- V5 : $\alpha(x + y) = \alpha \cdot x + \alpha \cdot y$
- V6 : $(\alpha + \beta)x = \alpha x + \beta x$
- V7 : $(\alpha\beta)x = \alpha(\beta x)$
- V8 : $1 \cdot x = x$

S 4.1:

- i : $0 \cdot x = 0$
- ii : $0 = 0$
- iii : $\alpha \cdot x = 0 \rightarrow x = 0 \vee \alpha = 0$
- iv : $(-\alpha)x = \alpha(-x) = -(\alpha \cdot x)$

Def polynomial space: $\mathcal{P}_n$ is defined as all polynomials of degree n . Further: $\mathcal{P} = \bigcup_{n=0}^{\infty} \mathcal{P}_n$

S 4.1: I a vectorspace the following holds for a scalar α and $x \in V$:

- $0x = 0$
- $0\alpha = 0$
- $\alpha x = 0 \Rightarrow \alpha = 0$ or $x = 0$
- $(-\alpha)x = \alpha(-x) = -(\alpha x)$

S 4.12: $\{b_1, \dots, b_n\} \subset V$ is a basis of $V \Leftrightarrow$ every vector $x \in V$ can be uniquely represented as: $x = \sum_{k=1}^n \xi_k b_k$

S 4.2: $\forall x \in V, \forall y \in V \exists z \in V : x + z = y$ where z is unique and $z = y - x$

Subspace

Def : A subspace (unterraum) U is a non-empty subset of V . It is closed under vector addition and scalar multiplication. U contains the zero-vector

S 4.3: Every subspace is a vectorspace
Def spanning set: The vectors $v_1, \dots, v_n$ are a spanning set (erzeugendes System) of V , if $\forall w \in \text{span}\{v_1, \dots, v_n\}$

Linear dependency, basis, dimensions

Def linear dependency: Vectors $v_1, \dots, v_n$ are linearly dependent

$\Leftrightarrow \sum_{k=1}^n \alpha_k \cdot v_k = 0 \rightarrow \alpha_1 = \dots = \alpha_n = 0$

Def dimension: the dimension of V is $\dim V = |\text{span} V|$ ($\dim\{0\} = 0$)

Lem 4.8: Every set $\{v_1, \dots, v_m\} \subset V$ with $|\mathbb{B}_v| < m$ is linear dependent

Cor 4.10: in a finite vectorspace, a set with n independent vectors is basis of V if $\dim(V) = n$

Def : The coefficients ξ_k are coordinates of x with respect to a basis $\mathbb{B} \xi = (\xi_1, \dots, \xi_n)^T$ is a coordinate vector

Def : Two subspaces $U, U' \subset V$ are complementary if every $v \in V$ has a unique representation in $U \cup U'$. Namely, $v = u \in U + u' \in U' \rightarrow V = U \oplus U'$

Linear Maps

Definitions

Def linearity: $F : V \rightarrow W$ is linear:

- $F(v + w) = F(v) + F(w)$
- $\alpha F(v) = F(\alpha v)$

Def injective: $\forall x, x' \subset X : f(x) = f(x') \Leftrightarrow x = x'$

Def surjective: $\forall y \subset Y, \exists x \subset X, f(x) = y$

Def bijective: surjective and injective $\Leftrightarrow f^{-1}$ exists

Matrix representation

Let F be a linear map $X \rightarrow Y$. One can write $F(b_i) \in Y$ as a linear combination of the basis of Y : $F(b_i) = \sum_{k=1}^m a_{k,i} \cdot c_k$ **Def :** The matrix $A^{m \times n}$ with the elements $a_{k,i}$ is a matrix (Abbildungsmatrix) with respect X, Y
 $F(x) = y \Leftrightarrow A\xi = \eta$ [H]

$$\begin{array}{ccc} X & \xrightarrow{F} & Y \\ \downarrow k_x \uparrow k_x^{-1} & & \downarrow k_y \uparrow k_y^{-1} \\ \mathbb{E}^n & \xrightarrow{A} & \mathbb{E}^m \end{array}$$

Def isomorphism: F is bijective $\Leftrightarrow F$ is an isomorphism

Def automorphism: F is isomorphism and $X = Y \Leftrightarrow F$ is an automorphism

S 5.1: F is isomorphism $\Leftrightarrow F^{-1}$ exists and is an isomorphism and linear

Kernel, Image and Rank

Def Kern: $\ker F = \{x \in X | F(x) = 0\}$

S 5.6: F injective $\Leftrightarrow \ker F = \{0\}$

Def Image: $\text{Im} F = \{F(x) | x \in X\}$

S 5.6: F surjective $\Leftrightarrow \text{im} F = Y$

ker A

is the solution set of $Ax = 0$. $\text{Im}(A)$ set of all b , such that $Ax = b$ is solvable **S 5.7:** $\dim X - \dim(\ker F) = \dim(\text{im} F) = \text{Rank}(F)$

Def : The rank F is equal to $\dim(\text{im}(F))$

Cor 5.8:

- $F : X \mapsto Y$ injective $\Leftrightarrow \text{Rank } F = \dim X$
- $F : X \mapsto Y$ surjective $\Leftrightarrow \text{Rank } F = \dim Y$

- $F : X \mapsto Y$ bijective (isomorphism) $\Leftrightarrow \text{Rank } F = \dim X = \dim Y$
- $F : X \mapsto Y$ bijective (automorphism) $\Leftrightarrow \text{Rank } F = \dim X, \ker F = \{0\}$

Cor 5.10:

- $\text{Rank}(G \circ F) \leq \min(\text{Rank } F, \text{Rank } G)$
- G is injective $\text{Rank}(G \circ F) = \text{Rank } F$
- G is surjective $\text{Rank}(G \circ F) = \text{Rank } G$

Matrices as linear mapping

Def columnspace: The columnspace (Spaltenraum) of A is the subspace $\mathcal{R}(A) = \text{im}(A) = \text{span}\{a_1, \dots, a_n\}$

Def nullspace: The nullspace (Nullraum) of A is a subspace $\mathcal{N}(A) = \ker A = L_0(Ax = 0)$

Def : # free variables = $\dim \mathcal{N}(A)$

S 5.12: Rank $A = r$: and L_0 Solution of $Ax = 0 \Rightarrow \dim L_0 = \dim \mathcal{N}(A) = \dim(\ker A) = n - r$

S 5.13: Rank $A \in M^{m \times n}$:

- pivots in Row-echelon-form
- $\dim(\text{im}(A))$ of $A : \mathbb{E}^n \mapsto \mathbb{E}^m$
- dimension of the linear independent columns/rows

Cor 5.14: $\text{Rank } A^T = \text{Rank } A^H = \text{Rank } A$

S 5.16: for $A \in \mathbb{E}^{m \times n}$ and $B \in \mathbb{E}^{p \times m}$:

- $\text{Rank } BA \leq \min(\text{Rank } A, \text{Rank } B)$
- $\text{Rank } B = m \leq p \Rightarrow \text{Rank } BA = \text{Rank } A$
- $\text{Rank } A = m \leq n \Rightarrow \text{Rank } BA = \text{Rank } B$

Cor 5.17: From S.5.16 it follows for quadratic matrices. $A \in \mathbb{E}^{m \times m}$ and $B \in \mathbb{E}^{m \times m}$

- $\text{Rank } BA \leq \min(\text{Rank } A, \text{Rank } B)$
- $\text{Rank } B = m \Rightarrow \text{Rank } BA = \text{Rank } A$
- $\text{Rank } A = m \Rightarrow \text{Rank } BA = \text{Rank } B$

S 5.18: For quadratic matrix $\mathbb{E}^{n \times n}$ the following statements are equivalent:

- A is regular
- $\text{Rank } A = n$
- Columns are linearly independent
- Rows are linearly independent
- $\ker A = \mathcal{N}(A) = \{0\}$
- A is invertible
- $\text{Im } A = \mathcal{R}(A) = \mathbb{E}^n$

S 5.19: For $Ax = b, b \neq 0$ with the solution x_0 and L_0 the solutionset is defined by $L_b = x_0 + L_0$ and is called **affine subspace** (not a real subspace since $0 \notin L_b$)
 $\dim(\text{Im}(A)) = n - \dim(\ker(A)) = n - (n - r) = r$

RC Find Basis of Im $A = \mathcal{R}(A)$:

- 1 bring into row echelon form
- 2 mark rows with pivots
- 3 marked columns in the normal form are a Basis

ex

$$\begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 3 & 0 & -6 & 0 & 0 & -12 & 15 & 9 \\ -3 & 4 & 1 & -3 & 0 & 0 & 0 & 0 \\ 1 & -4 & 3 & 3 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(1)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 3 & 0 & -6 & 0 & 0 & -12 & 15 & 9 \\ -3 & 4 & 1 & -3 & 0 & 0 & 0 & 0 \\ 1 & -4 & 3 & 3 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(2)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 0 & -12 & 15 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(3)} \text{span} \left\{ \begin{array}{c} -1 \\ 3 \\ -3 \\ 1 \\ -4 \end{array} \right\}$$

RC Find Basis of $\ker A = \mathcal{N}(A), A \in \mathbb{E}^{m \times n}$:

- 1 bring into row echelon form
- 2 create a vector for every row, which does not have a pivot. The dimensions of the vectors are $\mathbb{E}^{1 \times n}$ [3] Solve SLE $Ax = 0$ with the yielded vectors.
- 4 Write the solution as vector

ex

$$\begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 3 & 0 & -6 & 0 & 0 & -12 & 15 & 9 \\ -3 & 4 & 1 & -3 & 0 & 0 & 0 & 0 \\ 1 & -4 & 3 & 3 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(1)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 3 & 0 & -6 & 0 & 0 & -12 & 15 & 9 \\ -3 & 4 & 1 & -3 & 0 & 0 & 0 & 0 \\ 1 & -4 & 3 & 3 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(2)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 0 & -12 & 15 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(3)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 0 & -12 & 15 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \xrightarrow{(4)} \begin{array}{ccc|ccc} -1 & -4 & 7 & 3 & -1 & -4 & 7 & 3 \\ 0 & -12 & 15 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$x_4 = \alpha$
 $x_3 = \beta$
 $x_2 = \frac{5\beta + 3\alpha}{4}$
 $x_1 = 2\alpha$

span $\left\{ \begin{array}{c} 2 \\ 2 \\ 4 \\ 1 \\ 0 \\ 1 \end{array} \right\}$

Def Maps: Let X, Y be vector spaces with $\dim X = n, \dim Y = m$

- $F : X \mapsto Y$ a linear map
- $A : \mathbb{E}^n \mapsto \mathbb{E}^m, \xi \mapsto \eta^*$
- $B : \mathbb{E}^n \mapsto \mathbb{E}^m, \xi' \mapsto \eta'^*$
- $T : \mathbb{E}^n \mapsto \mathbb{E}^n, \xi \mapsto \xi'^*$
- $S : \mathbb{E}^m \mapsto \mathbb{E}^m, \eta \mapsto \eta'^*$

***Abbildungsmatrix**

****Transformationsmatrix**

$$\begin{array}{ccc} x \in X & \xrightarrow{F} & y \in Y \\ k_x \downarrow \uparrow k_x^{-1} & & k_y \downarrow \uparrow k_y^{-1} \\ \xi \in \mathbb{E}^n & \xrightarrow{A} & \eta \in \mathbb{E}^m \\ T^{-1} \downarrow \uparrow T & & S^{-1} \downarrow \uparrow S \\ \xi' \in \mathbb{E}^n & \xrightarrow{B} & \eta' \in \mathbb{E}^m \end{array}$$

$A = S B T^{-1}$
 $B = S^{-1} A T$

Wenn $F : X \rightarrow Y$,
 $A = T B T^{-1}$
 $B = T^{-1} A T$

S 5.20: Rank $F = r$ has the mappingmatrix*

$$A = \left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right]$$

Vector spaces with scalar products

Definitions

Def Norm: A norm is a function

$\|\cdot\| : V \rightarrow \mathbb{R}, x \mapsto \|x\|$ in a vector space which satisfies:

- N1 $\|x\| > 0, \|x\| = 0 \Leftrightarrow x = 0$ (positiv definit)
- N2 $\|\alpha x\| = |\alpha| \|x\|$ (homogenous)
- N3 $\|x \pm y\| \leq \|x\| + \|y\|$ (Triangle-inequality)

A normed vector space

has a norm **Def scalar product:** is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{E}, x, y \mapsto \langle x, y \rangle$, which satisfies:

- S1 $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ (linear in 2nd factor)
- S1 $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$ (linear in 2nd factor)
- S2 $\langle x, y \rangle = \langle y, x \rangle$ (symmetric, hermitian)
- S3 $\langle x, x \rangle > 0, \langle x, x \rangle = 0 \Leftrightarrow x = 0$ (positiv definit)

Def unit sphere: the set $\{x \in V | \|x\| = 1\}$

Def induced norm: The length of a vector is defined as: $\|\cdot\| : V \mapsto \mathbb{R}, \|x\| \mapsto \sqrt{\langle x, x \rangle}$

Def angle ϕ : $\phi = \angle(x, y), 0 \leq \phi \leq \pi$ is defined

by: $\phi = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{\mathcal{R}(x, y)}{\|x\| \|y\|}$

Def orthogonal vectors: two vectors x, y are orthogonal $\Leftrightarrow \langle x, y \rangle = 0$

Def orthogonal sets : two sets X, Y are orthogonal $\Leftrightarrow \forall x \in X, \forall y \in Y \langle x, y \rangle = 0$

S 6.1: $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle = \|x\|^2 \cdot \|y\|^2$ (Cauchy Schwarz Inequality)
S 6.2: $\|x \pm y\|^2 = \|x\|^2 + \|y\|^2 \Leftrightarrow x \perp y$ (Pythagoras)
Def orthogonal basis: $\Leftrightarrow \forall i, \forall j, i \neq j : \langle b_i, b_j \rangle = 0$
Def orthonormal basis: $\Leftrightarrow$ orthogonal basis with vectors of length 1
S 6.3: A set M of pairwise orthogonal vectors are linearly independent if $0 \notin M$
S 6.4: Let $\{b_1, \dots, b_n\}$ a orthonormal basis, $x \in V: x = \sum_{k=1}^n \langle b_k, x \rangle b_k \rightarrow \xi_k = \langle b_i, x \rangle$
S 6.5: from $\xi_k = \langle b_i, x \rangle_v, \eta_k = \langle b_l, x \rangle_v$ follows $\langle x, y \rangle_v = \sum_{k=1}^n \xi_k \eta_k = \xi^H \eta = \langle \xi, \eta \rangle_{\mathbb{R}^n}$ Which implies that if a basis in V is orthonormal the scalar product is valid in V
From that follows:
 $\|x\|_v = \|\xi\|_{\mathbb{R}^n}, \angle(x, y)_v = \angle(\xi, \eta)_{\mathbb{R}^n}, x \perp y \Leftrightarrow \xi \perp \eta$

RC Gram-Schmidt:

- $b_1 = \frac{a_1}{\|a_1\|_v}$
 - $\tilde{b}_k = a_k - \sum_{j=1}^{k-1} \langle b_j, a_k \rangle_v \cdot b_j$
 - $b_k = \frac{\tilde{b}_k}{\|\tilde{b}_k\|_v}$
- ex**
- $$A = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \rightarrow a_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} / \left\| \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\| = \frac{2/3}{1/3} \tilde{a}_2 = \frac{3}{1} - \left\langle \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\rangle = \frac{-1/3}{2/3}, a_2 = \frac{-1/3}{2/3} / \left\| \begin{pmatrix} -1/3 \\ 2/3 \end{pmatrix} \right\| = \frac{-1/3}{-2/3} \rightarrow \begin{pmatrix} 2/3 & -1/3 \\ 2/3 & -2/3 \end{pmatrix}$$

S 6.6: After k-steps the set $\{b_1, \dots, b_k\}$ is pairwise orthonormal. $\{b_1, \dots, b_k\}$ is a basis $\Leftrightarrow \{a_1, \dots, a_k\}$ is a basis Every vectorspace ($\neq \infty$) has a orthonormal basis
Cor 6.7: To a vector space with scalar product with finite or countably infinite many dimensions a orthonormal basis exists.
Def orthogonal complement: $U^\perp$ is the orthogonal complement of a subspace U .
 $U \oplus U^\perp = V$

S 6.9: For a complex matrix with $rank A = r$ it holds:

- $N(A) = \mathcal{R}(A^H)^\perp \subset \mathbb{C}^n$
- $N(A^H) = \mathcal{R}(A)^\perp \subset \mathbb{C}^m$
- $N(A) \oplus \mathcal{R}(A^H) = \mathbb{C}^n$
- $N(A^H) \oplus \mathcal{R}(A) = \mathbb{C}^m$
- $dim \mathcal{R}(A) = r$
- $dim \mathcal{R}(A^H) = r$
- $dim N(A) = n - r$
- $dim N(A^H) = m - r$

Those are the **fundamental subspaces**

Change of Basis

B, B' are orthonormalbasis. Hence:
 $b'_k = \sum_{j=1}^n T_{jk} b_j$ Matrix for change of basis T :
 $T^{-1} = T^H$ since both basis are orthonormal.
Therefore it holds that:

- $\xi = T \xi'$
- $\xi' = T^{-1} \xi$
- $B = B' T$
- $B' = B T^H$

S 4.13: Let $\xi = (\xi_1 \dots \xi_n)^T$ be a coordinate vector of an arbitrary vector $v \in V$ with respect to the **old basis**
. Let $\xi' = (\xi'_1 \dots \xi'_n)^T$ be the new representation of a vector x with respect to the new basis.
 $x = \sum_{i=1}^n \xi_i b_i = \sum_{k=1}^n \xi'_k b'_k$

All matrices are unitary/orthogonal **Cor 6.12:**
 $\langle x, y \rangle_v = \xi^H \eta = \langle \xi, \eta \rangle_v = \langle \xi', \eta' \rangle_v = \xi'^H \eta'^T \Rightarrow T$ is conformal (längen-winkeltreu)
Note Convention: A representation of a vector with respect to the basis $\mathcal{B}_1$ is written as $[v]_{\mathcal{B}_1}$
Therefore: $[v]_{\mathcal{B}_2} = Mat(\mathcal{B}_1)_{\mathcal{B}_2} [v]_{\mathcal{B}_1}$ and $[v]_{\mathcal{B}_1} = Mat(\mathcal{B}_2)_{\mathcal{B}_1} [v]_{\mathcal{B}_2}$ where $Mat(\mathcal{B}_2)_{\mathcal{B}_1}$ is the matrix of change ob basis from $\mathcal{B}_1$ to $\mathcal{B}_2$ Hence: $Mat(\mathcal{B}_1)_{\mathcal{B}_2} = ([b_1]_{\mathcal{B}_2} \dots [b_n]_{\mathcal{B}_2})$

RC Calculating the matrix of F with respect to Basis $\mathcal{B}$: We have a function $F: X \mapsto X$ and the basis of $X := \mathcal{B}$:

- calculate for all $\forall a \in \mathcal{B}: F(b_a)$
- solve for all $F(b_a) = \alpha_a b_1 + \beta_a b_2 \dots \gamma_a b_n$
- write coordinate vectors as: $\xi_a = (\alpha_a \beta_a \dots \gamma_a)^T$
- write matrix as $F_{[\mathcal{B}]} = \xi_1 | \dots | \xi_n$

RC Compute Basistransformationsmatrix from $\mathcal{B}$ to $\mathcal{S}$:

Since $\mathcal{S}$ is a standardbasis we have:

- $\mathcal{S} \rightarrow \mathcal{B}$ is given by $B = (s_1 | s_2 | \dots | s_n)$ (Columns of B are the basis vectors of $\mathcal{B}$)
- Compute inverse of B to get $\mathcal{B} \rightarrow \mathcal{S}$

RC Basistransformationsmatrix from 2×2 matrices with respect to the standard basis:

The standard basis of 2×2 matrices is given by: $\mathcal{S} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ and the new Basis is defined by: $\mathcal{B} = \left\{ \begin{pmatrix} a & b \\ c & e \end{pmatrix}, \begin{pmatrix} e & f \\ g & h \end{pmatrix}, \begin{pmatrix} i & j \\ k & l \end{pmatrix}, \begin{pmatrix} m & n \\ o & p \end{pmatrix} \right\}$ Then the matrix (Abbildungsmatrix) is defined by $F = \begin{pmatrix} a & e & i & m \\ b & f & j & n \\ c & g & k & o \\ d & h & l & p \end{pmatrix}$

RC Prove that $\mathcal{B}$ is Basis:

$\mathcal{S}$ denotes the standard basis.

- compute for all $\forall a \in \mathcal{B}: b_a = \alpha s_1 + \beta s_2 \dots \gamma s_n$
- Since $span(\mathcal{B}) = span(\mathcal{S})$, $\mathcal{B}$ has to be basis.

RC Prove that F is a bijective mapping

$F: X \mapsto Y$ with the basis $\mathcal{X}$ for X and $\mathcal{Y}$ for Y :

- calculate for all $\forall a \in \mathcal{X}: F(x_a)$
- solve for all $F(x_a) = \alpha_a y_1 + \beta_a y_2 \dots \gamma_a y_n$
- write coordinate vectors as: $\xi_a = (\alpha_a \beta_a \dots \gamma_a)^T$
- write matrix as $F_{[\mathcal{X}]} = \xi_1 | \dots | \xi_n$
- As $F_{[\mathcal{X}]}$ is quadratic and has full rank we have that $dim(\mathcal{X}) = dim(\mathcal{Y})$ and thus by Cor.5.8 that F is bijective

unitary/orthogonal mapping

Def unitary: A linear mapping $F: X \mapsto Y$ is unitary/orthogonal if $\langle F(v), F(w) \rangle_y = \langle v, w \rangle_x$

S 6.13:

- F is isometric (längentreu): $\|F(v)\|_y = \|v\|_x$
- F is conformal (winkeltreu): $v \perp w \Leftrightarrow F(v) \perp F(w)$
- $ker F = \{0\}$, F is injective

- if $n = dim X = dim Y < \infty$
- F is isomorphism
- $\{b_1, \dots, b_n\}$ is orthonormal basis of $X \Leftrightarrow \{F(b_1), \dots, F(b_n)\}$ is a orthonormal basis of Y
- F^{-1} is unitary/orthogonal
- The mapping matrix (Abbildungsmatrix) A is unitary/orthogonal

Least Squares

Let $Ax = b$ a overdetermined SLE (Equations & Variables). No exact solution exists.
 $\rightarrow x^* = argmin_{x \in \mathbb{R}^n} \|Ax - b\|_2^2 \Rightarrow (Ax - b) \perp \mathcal{R}(A)$

Def Pseudoinverse: If Rank $A = n$:
 $A^+ = (A^H A)^{-1} A^H \Rightarrow A^+ A = I$
Def normalequations: $(A^T A)x = A^T y$

RC Least Square Method for functions:

We assume that $ker(A) = \{0\}$ and $A^H A$ is regular

- bring problem in a form where everything is numerically determined except the coefficients
- calculate $A^T y$
- calculate $A^T A$
- solve the equation $(A^T A)x = A^T y$
- calculate error $r = y - Ax$

ex

The equation is given $y(t) = x_1 t + x_2 t^2$ We have $t_n(1, 2, 3, 4)$ and $y(t)_n = (13.0, 35.5, 68.0, 110.5)$

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \end{pmatrix}, y = \begin{pmatrix} 13.0 \\ 35.5 \\ 68.0 \\ 110.5 \end{pmatrix} \xrightarrow{1} A^T y = \frac{730.0}{2535.0} \xrightarrow{2} A^T A = \begin{pmatrix} 30 & 100 \\ 100 & 354 \end{pmatrix} \xrightarrow{3} \begin{pmatrix} 30 & 100 \\ 100 & 354 \end{pmatrix} \begin{pmatrix} 730 \\ 2535.0 \end{pmatrix} \mapsto x_1 = 7.9355, x_2 = 4.919$$

RC Least Square Method for 2D-points:

- Write X- / Y-coordinate alternately in the form $(x, 0), (0, y)$ in A for every point. Write X- / Y-coordinate alternately in y .
- Rest as usually

ex

The points $P = \{(-1, 1), (1, 1), (1, -1), (-1, -1)\}$ should be transformed with respect to the squared distance to the points $P' = \{(0, 2), (1, 3), (0, -2), (-1, -3)\}$. The transformation is defined as $T(P) = T \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} s_x \cdot p_x \\ s_y \cdot p_y \end{pmatrix}$

$$Ax = y \Rightarrow \begin{pmatrix} -1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} s_x \\ s_y \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 3 \\ 0 \\ -2 \\ -1 \\ -3 \end{pmatrix}$$

RC Least Squares with QR-decomposition:
We have the normal equations $A^H Ax = A^T b \Rightarrow (QR)^H (QR)x = (QR)^H b \Rightarrow R^H Q^H QRx = R^H Q^H b \Rightarrow R^H Rx = R^T Q^T b \Rightarrow Rx = Q^H b$ Therefore we have:

- compute QR-decomposition of A
- solve $Rx = Q^T b$

RC Least Squares with SVD:

$$\|Ax - b\|_2^2 = \left\| \underbrace{\Sigma V^H}_y x - \underbrace{U^H b}_c \right\|_2^2 = \|\Sigma y - c\|_2^2; x^* = V \Sigma^+ U^H b \Rightarrow \infty \text{ solutions, here: smallest 2-norm } (y^* = \Sigma^+ U^H b) \text{ Where } \Sigma^+ \text{ is the pseudoinverse of } \Sigma, \text{ hence it holds}$$

Determinants

Def :

$det(a_{11}) = a_{11}, det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21},$
 $det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = +a_{31}a_{12}a_{23} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$
S 8.12: $det(A) = \sum_{i=1}^n a_{ki} K_{ki} = \sum_{i=1}^n a_{il} K_{il}$ for a fixed k and l .

Def cofactor K_{ki} : $K_{ki} = (-1)^{k+i} det(A_{[k,i]})$
Def $A_{[k,i]}$: Is defined as the matrix A without the k -th row and i -th column
Def : $detA = 0 \Leftrightarrow A$ is singular
Def : $detA \neq 0 \Leftrightarrow A$ is regular

S 8.3:

- $det(A)$ is linear in every row
 - swapping two rows changes the sign of $det(A)$
 - $det(I) = 1$
- S 8.4:**
- if A has a row with 0 $\Rightarrow det(A) = 0$
 - $det(\gamma A) = \gamma^n det(A)$
 - if A has to equal rows $\Rightarrow det(A) = 0$
 - adding a multiple of a row to another row doesn't affect the det
 - is A a diagonal matrix: $det(A) = \prod_{i=1}^n a_{ii}$
 - is A a triangular matrix: $det(A) = \prod_{i=1}^n a_{ii}$

Cor 8.10: Every statement of Satz 8.3 and 8.4 also holds for columns instead of rows

S 8.5: Using Gauss on A results in:
 $det(A) = (-1)^v \prod_{k=1}^n r_{kk}$ where v is # swappings of rows and r_{kk} are the diagonal elements of the row echelon form **S 8.7:** $det(AB) = det(A) \cdot det(B)$

Cor 8.8: if A is regular $\Rightarrow det(A^{-1}) = \frac{1}{det(A)}$

S 8.9: $det(A^T) = det(A)$ and $det(A^H) = \overline{det(A)}$
Def det of block matrices:

$det \left[\begin{array}{c|c} A & C \\ \hline 0 & B \end{array} \right] = det(A) \cdot det(B)$
Def det of unitary matrices: Let A be unitary/orthogonal $|det(A)| = \pm 1$ proof:
 $det(U^T U) = det(I) = det(UU^T) = 1 \Rightarrow det(U) = \pm 1$

Eigenvalues and -vectors

Def eigenvector: A number $\lambda \in \mathbb{R}^n$ is called eigenwert of a linear mapping: $F: X$ if $\exists v \in V, v \neq 0$ such that $F(v) = \lambda v$. v is an eigenvector. The set of all eigenvectors, which correspond to λ form a subspace $E_\lambda = \{v \in V | F(v) = \lambda v\}$
Def spectrum: The set of all eigenvalues of F is called spectrum
Def : $\xi \in \mathbb{R}^n$ is a eigenvector of $\lambda \Leftrightarrow A\xi = \lambda\xi$
Lem 9.1: A linear map F and its matrix representation have the same eigenvalues and the eigenvectors are connected by the coordinaterepresentation k_v
Lem 9.2: λ is eigenvalue $\Leftrightarrow ker(A - \lambda I)$ is singular ($E_\lambda = ker(A - \lambda I)$)
Def multiplicity: The geometric multiplicity of $\lambda = dim(E_\lambda)$

Def characteristic polynomial: It is defined by $\chi_A(\lambda) = \det(A - \lambda I) = 0$
Def Trace: $tr(A) = \sum_{k=1}^n a_{kk}$
S 9.5: $\lambda \in \mathbb{E}$ is eigenvalue of **A** $\Leftrightarrow \chi_A(\lambda) = 0$
Lem 9.4:
 $\chi_A(\lambda) = (-1)^n \cdot \lambda^n (-1)^{n-1} \cdot tr(A) \cdot \lambda^{n-1} + \dots \det(A) \lambda^0 = a_n \cdot \lambda^n + a_{n-1} \cdot \lambda^{n-1} + \dots \det(A)$
Lem 9.6: A (quadratic) matrix is singular if and only if it has 0 as an eigenvalue
Def algebraic multiplicity: is the multiplicity of an eigenvalue in the char. polynomial.
S 9.13: geometric multiplicity $\leq$ algebraic multiplicity

RC Find Eigenvalues and -vectors:
1 find char. polynomial $\chi_A(\lambda) = \det(A - \lambda I)$
2 find roots of χ_A
3 for every λ_k find the solution for $(A - \lambda_k I)x = 0$

S 9.7: for similar matrices $C = T^{-1}AT$ (C and A are similar) holds that
 $tr(A) = tr(C)$, $\det(A) = \det(C)$, $\chi_A = \chi_C$ and they have the same eigenvalues
S 9.11: Eigenvectors for different Eigenvalues are linearly independant $\Rightarrow \max \dim V$ different Eigenvalues
Def : λ is eigenvalue for $A \Rightarrow \lambda^q$ is eigenvalue of A^q
Note Trace: $trace(A)$ is equal to the sum of all eigenvalues of A : $trace(A) = \lambda_1 + \dots + \lambda_n$
Note Trace: $\det(A)$ is equal to the product of all eigenvalues of A : $\det(A) = \lambda_1 \cdot \dots \cdot \lambda_n$

Decompositions

Spectral-/Eigenvaluedecomposition

Def : $A = V\Lambda V^{-1}$ (A and Λ are similar)
Precondition for diagonalisation
: **S 9.14:** $A \in \mathbb{C}^{n \times n}$ is diagonalisable $\Leftrightarrow \forall$ Eigenvalues (geom. mult. = alg. mult.)
S 9.15: If $A \in \mathbb{C}^{n \times n}$ is unitary ($A^H = A$) it holds that:
i all eigenvalues are real
ii the eigenvectors are pairwise orthogonal
iii an orthonormal basis U exists, which consists of all the eigenvectors
iv for the unitary matrix U holds that $U^H AU = \Lambda$
Cor 9.16: The previous statements is also valid for real-symmetric matrices

RC Eigenvaluedecomposition:
1 find the eigenvalues of **A** λ_k and write $\Lambda = \text{diag}(\lambda_1 \dots \lambda_n)$
2 find the according eigenvectors v_k of λ_k write them as $V = (v_1 | \dots | v_n)$ (sorted according to Λ)
3 find inverse V^{-1}

Cor 9.10: If A is diagonalisable it can be composed as a sum of **1-rank-matrices**
: $A = \sum_{k=1}^n V_k \lambda_k w_k^T$ with $V = (V_1 \dots V_n)$ and $V^{-1} = \begin{pmatrix} \vdots \\ w_n^T \end{pmatrix}$ from that follows: $Av_k = v_k \lambda_k$
and $w_k^T A = \lambda_k w_k^T$: w^T is a **left eigenvector**

RC Eigenvaluedecomposition with SVD:
The SVD is given by $A = U\Sigma V^H$
1 Expand $U\Sigma V^H \Rightarrow U I \Sigma V^H \Rightarrow U I_1 I_2 \Sigma V^H$ where $U I_1 = V$ and $I_1 I_2 = I$
2 Calculate $(U I_1) (I_2 \Sigma) V^H$

RC Composition of 1-rank- matrices:
1 write $A = V\Lambda V^{-1}$
2 rewrite $A = \sum_{k=1} V_k \lambda_k w_k^T V_k = \text{row} \downarrow$ of V , $w_k^T = \text{column} \rightarrow$ of V^{-1}

RC Powers of A:
1 write $A = V\Lambda V^{-1}$
2 calculate $A^m = V\Lambda^m V^{-1}$
note: $\Lambda^m = \text{diag}(a_{11}^m, \dots, a_{nn}^m)$

Singularvaluedecomposition

Def SVD for $A^H A$: Spectral-decomposition exists for every matrix $A^H A$. Since $A^H A$ is hermetian (hermetisch) and positive semidefinite. $\rightarrow A^H A$ has real, non-negative eigenvalues $\lambda \in \mathbb{R}$ and $\lambda \geq 0$ Therefore one can rewrite: $A^H A V = V \Lambda \xrightarrow{\lambda=\sigma^2} A^H A V_r = V_r \Sigma_r^2 \Rightarrow V_r^H A^H A V_r = \Sigma_r^2 \Rightarrow (\underbrace{\Sigma_r^{-1} V_r^H A^H}_{=U_r^{-1}}) (\underbrace{A V_r}_{=U_r}) = I$

Def SVD: SVD exists for every matrix, such that U, V are unitary and Σ is diagonal and positive. $A = U\Sigma V^H$ it follows $AA^H = U\Sigma_m^2 U^H, A^H A = V\Sigma_n^2 V^H, A^H = V\Sigma^T U^H$
Def A invertible: If A is invertible $A^{-1} = V\Sigma^{-1} U^H$
S 11.11f: For Rank r it holds that:

- $\{u_1, \dots, u_1\}$: Basis of $Im(A) = \mathcal{R}(A)$
- $\{u_{r+1}, \dots, u_m\}$: Basis of $Ker(A^H) = \mathcal{N}(A^H)$
- $\{v_1, \dots, v_1\}$: Basis of $Im(A^H) = \mathcal{R}(A^H)$
- $\{v_{r+1}, \dots, v_m\}$: Basis of $Ker(A) = \mathcal{N}(A)$

Def Automorphism: In a self-image (Selbstabbildung) it holds that
 $A = U\Sigma V^H = V \underbrace{V^H U}_{R} \Sigma \underbrace{V^H}_{\substack{1 \quad 2 \quad 3 \quad 4}} =$
1,4 Change to orthonormal basis
2 rotation, mirroring
3 scaling of unit-axes
Def singular values: singular values: $\sigma_i = \sqrt{\lambda_i}$ sorted in descending order $\sigma_a \leq \sigma_b \dots \sigma_r \leq 0 \dots$
Def Eigenbasis: V is the orthonormal eigenbasis of $A^H A$ such that: $AA^H = U\Sigma_m^2 U^H$.
Similar for U as eigenbasis of AA^H :
 $A^H A = V\Sigma_n^2 V^H$

RC SVD of $A \in \mathbb{E}^{m \times n}$ with $A^H A$:
1 Calculate $(A^H A) \in \mathbb{E}^{n \times n}$
2 find eigenvalue of $A^H A$
3 write $\Sigma_r = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \in \mathbb{E}^{n \times n}$
4 rewrite: $\Sigma \in \mathbb{E}^{m \times n}$: $\Sigma = \begin{smallmatrix} \Sigma_r & 0 \\ 0 & 0 \end{smallmatrix}$
5 find eigenvectors of $A^H A \Rightarrow v_1, \dots, v_r$
6 norm eigenvectors and compute $V = (\frac{v_1}{\|v_1\|} | \dots | \frac{v_n}{\|v_n\|}) \in \mathbb{R}^{n \times n}$
7 solve for $U = AV\Sigma^{-1}$
8 if $U_r \neq U$: $U_r \xrightarrow[\text{gram}]{schmidt} U \in \mathbb{E}^{m \times m}$
9 write $A = U\Sigma V^H$

RC SVD of $A \in \mathbb{E}^{m \times n}$ with AA^H :
1 Calculate $(AA^H) \in \mathbb{E}^{m \times m}$
2 find eigenvalue of AA^H
3 write $\Sigma_r = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \in \mathbb{E}^{m \times m}$
4 rewrite: $\Sigma \in \mathbb{E}^{m \times n}$: $\Sigma = \begin{smallmatrix} \Sigma_r & 0 \\ 0 & 0 \end{smallmatrix}$
5 find eigenvectors of $AA^H \Rightarrow v_1, \dots, v_r$
6 norm eigenvectors and compute $V = (\frac{v_1}{\|v_1\|} | \dots | \frac{v_n}{\|v_n\|}) \in \mathbb{R}^{n \times n}$
7 solve for $U = AV\Sigma^{-1}$
8 if $U_r \neq U$: $U_r \xrightarrow[\text{gram}]{schmidt} U \in \mathbb{E}^{m \times m}$
9 write $A = U\Sigma V^H$

RC SVD with Spectral decomposition
 $A = V\Lambda V^{-1}$:
One has to sort the singular values of Λ according to their value. Then one can do:
1 We have $U = V, \Sigma = \sqrt{\Lambda}$
2 Rewrite: $A = V\sqrt{\Lambda} V^{-1}$

Cor 11.4: If $A \in \mathbb{E}^{m \times n}$ and $rank(A) = r$ then: eigenvalues of $A^H A \in \mathbb{E}^{m \times m}$ and $AA^H \in \mathbb{E}^{n \times n}$ are the same but the multiplicity of the eigenvalue 0 is $n - r$ or $m - r$
Def spectral norm: It is defined as: $\|A\|_2 = \sigma_1$

QR-Decomposition

Def QR-Decomposition: A matrix **A** can be composed as $A = QR$ where Q is orthohogonal and R is an upper triangular matrix. The decomposition is unique if $m \leq n$ and $Rank(A) = n$

RC QR-Decomposition:
1 Gram Schmidt on the rows ($\downarrow$) of $A \rightarrow Q$
2 solve $R = Q^T A \rightarrow R$

ex
 $A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{1} q_1 = \begin{pmatrix} 2/3 \\ 2/3 \\ 1/3 \end{pmatrix}, q_2 = \begin{pmatrix} -1/3 \\ 2/3 \\ -2/3 \end{pmatrix} \mapsto Q = \begin{pmatrix} 2/3 & -1/3 \\ 2/3 & 2/3 \\ 2/3 & -2/3 \end{pmatrix} \xrightarrow{2} R = Q^T A = \begin{pmatrix} 3 & 5 \\ 0 & 1 \end{pmatrix}$

Definitions

Def nullmatrix: Has in every entry 0
• $\forall A(A + 0 = 0 + A = A)$ (S.2.2)
Def diagonalmatrix: Has in every entry 0 except for the diagonal: $(D)_{ij} = 0$ for $i \neq j$ one can write
 $\text{Diag}(d_{11}, \dots, d_{nn})$
• is symmetric
• $\det(A) = A_{11} \cdot \dots \cdot A_{nn}$ (S.8.4)
• $A^{-1} = \text{diag}(\frac{1}{A_{11}}, \dots, \frac{1}{A_{nn}})$
• $A^m = \text{diag}(a_{11}^m, \dots, a_{nn}^m)$
Def identity: The identity is written as
 $I_n = \text{Diag}(1, \dots, 1)$
• $A \cdot I = A = I \cdot A$
• $A^{-1} = I$
Def upper triangular matrix: $(R)_{ij} = 0$ for $i > j$
• is nilpotent
Def lower triangular matrix: $(R)_{ij} = 0$ for $i < j$
• is nilpotent
Def Zerodiviser: If $AB = 0 \Leftrightarrow A, B$ Zerodiviser, Nullteiler
Def symmetric: $A^T = A \Leftrightarrow A$ symmetric
• A and B symmetric $\Rightarrow AB = BA \Leftrightarrow AB$ is symmetric (S.2.7)
• if positiv definit $\Rightarrow$ regular (L3.7)
Def skew-symmetric: $A^T = -A \Leftrightarrow A$ skew-symmetric
• $\text{tra}(A) = 0$
• if A has odd order
– $\det(A) = 0$
– do inverse exists
– A is singular

if A has even order
• inverse is skew-symmetric if it exists
Def hermitian: $A^H = A \Leftrightarrow A$ hermitian

- if positiv definit $\Rightarrow$ regular (L.3.7)
- Def unitary/orthogonal:** $A^H A = I \Leftrightarrow A$ is unitary/orthogonal
- A is regular (S.2.20)
 - $A^{-1} = A^H$ (S.2.20)
 - A^{-1} is unitary (S.2.20)
 - A and B is unitary $\Rightarrow AB$ is unitary (S.2.20)
 - $\det(A) = \pm 1$

Multiple Choice

General
C If the solutions of an SLE are $x_1 = 0, x_2 = 0, x_3 = 1$ the system has infinite many solutions
C Let A be a real 2×4 matrix with rank 2. Then the SLE $Ax = b$ has a non-trivial solution
W If A is invertible it holds: $ABA^{-1} = B$
C $\mathbb{E}^{n \times n} \rightarrow \mathbb{E}, A \mapsto \text{trace}(A)$ is linear
W $\mathbb{E}^{n \times n} \rightarrow \mathbb{E}, A \mapsto \det(A)$ is linear
C Let $D \in \mathbb{E}^{2 \times 2}$, $\dim(Ker D) = 2$ only if $D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
C If A^2 is invertible, so is A^3
 $\det(A^2) \neq 0 \Rightarrow \det(A) \neq 0 \Rightarrow \det(A^3) \neq 0$
C If A is regular and $A^2 = A$, then $A = I$
W For linear dependant x, y, z it holds $x = \alpha y + \beta z$
Only x, y can be dependant
C If A and $A^2 \in \mathbb{E}^{n \times n}$ and A^2 is regular, A^3 is invertible
C $\forall x \in \mathbb{R}^n, \|Ax\|_2 \leq \|A\|_2 \|x\|_2$
W Let $f: \mathbb{R}^n \rightarrow \mathbb{R}, f(x) := \|Ax\|_2$ The function f is a norm in $\mathbb{R}^n$
W $\|AB\|_2 \leq \|A\|_2$
Given are the orthogonal matrices A and B with the same dimension. Which of the following properties is true?
W The matrix product AB is orthogonal, but BA is not orthogonal
W The matrix product BA is orthogonal, but AB is not orthogonal.
C The matrix product AB and the matrix product BA are orthogonal
W The matrix product AB and the matrix product BA are not orthogonal It holds that $A^T = A^{-1}$
 $BT = B^{-1}$ and further $(AB)^T = B^T A^T = B^{-1} A^{-1} = (AB)^{-1}$ and also vice versa $(BA)^T = (BA)^{-1}$
We have $\mathbb{R}^n$ with the standard scalar product $\langle \cdot, \cdot \rangle$ and 2-norm. Let A be a real $n \times n$ matrix. Which statements are correct
C $\forall x, y \in \mathbb{R}^n$ it holds that $\langle x, ATy \rangle = \langle Ax, y \rangle$
C $A^T = A^{-1} \Rightarrow \forall x, y \in \mathbb{R}^n \langle Ax, Ay \rangle = \langle x, y \rangle$
C $A^T = A^{-1} \Rightarrow \forall x \in \mathbb{R}^n \|Ax\| = \|x\|$
C Let B be another real $n \times n$ matrix.
 $A^T = A^{-1}$ and $B^T = B^{-1} \Rightarrow$ the inverse of AB exists and it is: $(AB)^{-1} = (AB)^T$

Given are orthogonal matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$. Which of the following statements are correct?
C The matrix A^T is orthogonal
W The matrix $A + B$ is orthogonal
W The matrix $A + A^T$ is orthogonal
C The matrix AB^{-1} is orthogonal

Given is a lower triangular matrix $A \in \mathbb{R}^{3 \times 3}$ whose entries are non-negative integers and whose entries either occur only once or are equal to zero. Which of the following options are possible for the value of the determinant $\det(A)$?
W 5
W 7
W -2
C 35 A is triangular $\Rightarrow \det(A) = a_{1,1} a_{2,2} a_{3,3} \Rightarrow$ so $\det(A) = 0$ or $\det(A) = a_{1,1} a_{2,2} a_{3,3}$
The dimensions of the subspace of all skew-symmetric real 3×3 matrices is:
W 1
C 3
W 6
W 9
Let $A \in \mathbb{R}^{2 \times 3}$ and $b \in \mathbb{R}^2$. Assume a solution for $Ax = b$ exists
C $Ax = b$ has always ∞ solutions min 1 free variable
W The set of the solution (Lösungsmenge) of $Ax = b$ forms a line in 3D could also be a plane, 1 or 2 free variables
C Geometrically $Ax = b$ corresponds to an intersection of two planes in 3D

Rank

W Let $B \in \mathbb{E}^{3 \times 1}$ and $C \in \mathbb{E}^{1 \times 3}$; BC can have rank 3.

Vectorspaces

- W Let V be a vector space over $\mathbb{R}$ with scalar product $\langle \cdot, \cdot \rangle$ and let $F: V \mapsto V$ be a linear map. If it holds that $\forall v \in V, \langle v, F(v) \rangle = 0$, then F is necessarily the null map, i.e., $F(v) = 0$ for all $v \in V$.
- W Given a vector space V with a norm $\|\cdot\|$. For all $u, v \in V$, we have $\|v\| \leq \|v + u\|$.
counterexample: $u = -v = (1, 1)^T$
- C Let $S \subset V$ and W be a subspace of V :
 $S \subset W \Rightarrow \text{span}(S) \subset W$
- C In a vector space of finite dimension with scalar product, one can complete any set of orthonormal vectors to form an orthonormal basis.
- C A vector space of finite dimension with scalar product has an orthonormal basis. Corollary 6.7
- W Consider the vector space $\mathbb{R}^n$ with the Euclidean scalar product. The scalar product of two unit vectors can be arbitrarily large. Cauchy Schwarz, S 6.1: $\langle v, w \rangle^2 \leq \langle v, v \rangle \langle w, w \rangle = 1 \cdot 1 = 1$
- W We again consider $\mathbb{R}^n$ with the Euclidean scalar product. Can we find any number of pairwise orthogonal unit vectors in this vector space?

Let V the standard vectorspace of all 2×2 matrices.
Which of the following are subspaces of V ? ($B = \frac{1}{3} \begin{pmatrix} 2 & 4 \\ 3 & 4 \end{pmatrix}$)

W $\{A \in V | A \text{ is invertible}\}$

W $\{A \in V | A^2 = 0\}$

C $\{A \in V | A^T = A\}$

C $\{A \in V | A^T B = BA\}$

- Consider the vector space F of functions of $\mathbb{R} \mapsto \mathbb{R}$ with the operations addition $(f + g)(x) = f(x) + g(x)$ and scalar multiplication $(\lambda f)(x) = \lambda f(x)$. This includes the subspace $P_2 = \{a_0 + a_1x + a_2x^2 | a_i \in \mathbb{R}\}$ of polynomials of degree ≤ 2 . Which of the following statements are correct?
- C $\text{Span}\{x + 1, x^1, x^2 + 1, x^2\}$ is equal to P_2 .
- W $x + 1, x^1, x^2 + 1, x^2$ in P_2 are linearly independent
- W $x + 1, x^1, x^2 + 1, x^2$ in P_2 form a generating set (spanning set) of F .
- C $x + 1, x^1, x^2 + 1, x^2$ in P_2 form a generating set (spanning set) of P_2 .
- W The polynomials $x + 1, x^1, x^2 + 1, x^2$ in P_2 form a basis of P_2
- Let V, W be finite dimension vector spaces over a space K . Let $F: V \mapsto W$ be a linear mapping and $(v_1, \dots, v_n)$ a basis of V . Then it holds that:
- W $F(v_1), \dots, F(v_n)$ are linearly independent if F is surjective
- C $F(v_1), \dots, F(v_n)$ are linearly independent if F is injective
- C $F(v_1), \dots, F(v_n)$ form a generating end system if F is surjective
- W $F(v_1), \dots, F(v_n)$ form a generating end system if F is injective
- C $F(v_1), \dots, F(v_n)$ form a basis if and only if F is an isomorphism

- Let V, W be two real vector spaces with scalar products, let B be an orthonormal basis of V and let $F: V \mapsto W$ be an orthogonal mapping. Which of the following statements is true?
- W F is an isomorphism. F is only an isomorphism if $\dim V = \dim W < \infty$
- C $\|F(v)\|_W = \|v\|_v$ for all $v \in V$. Follows directly from the definition of the scalar product induced norm and the orthogonality of F .
- W If $\dim V, \dim W < \infty$, it is possible that $\dim V > \dim W$ With $\dim V > \dim W$ there is no orthogonal mapping $F: V \mapsto W$.
- W F is not injective.
- C is angle-preserving, i.e., for all $v, w \in V$ it holds that $\langle F(v), F(w) \rangle = \langle v, w \rangle$.
- C F is an isomorphism to the image of F . F is injective as shown before and obviously surjective to its image. Since the inverses of linear mappings are also linear, F is therefore an isomorphism
- W The set of images $F(B)$ is an orthonormal basis of W . Since F is not necessarily an isomorphism, the image space can be smaller than W .
- C The set of images $F(B)$ is an orthonormal basis of $\text{Im}(F)$ This follows from the isomorphism property of F from the question above
- C If it exists, $F^{-1}: \text{Im}(F) \mapsto V$ is orthogonal. It exists as seen above. The orthogonality of the

inverse follows directly from the definition of orthogonal mappings.

Det

- W Let Q be unitary and $A \in \mathbb{E}^{n \times n}$,
 $\det(QA) = \det(A) \quad |\det Q| = \pm 1$
- C If $A, B, P \in \mathbb{E}^{n \times n}$ and P is invertible with $A = PBP^{-1}$ then: $\det(A) = \det(B)$
- Given is a matrix $A \in \mathbb{R}^{n \times n}$ with entries $a_{ij} = ij$ and $n > 1$. Which statement is correct?
- W $\det(A) = 1$
- C $\det(A) = 0$
- W $\det(A) = (-1)^n$
- W $\det(A) = (-2)^n$
- Which of the following statements are not correct for arbitrary $n \times n$ -matrices A and B ?
- C $\det(A + B) = \det(A) + \det(B)$
- W $\det(AB) = \det(BA)$
- W If A is singular then AB is also singular
- W $\det(AA^T A) = (\det(A))^3$

- Let $A, B \in \mathbb{R}^{n \times n}$ with $AB = -BA$
- C $\det(AB) = \det(-BA)$
- W $\det(A)\det(B) = -\det(A)\det(B)$ n has to be even $\Rightarrow$ S8.4 v
- W Either A or B has a zero-determinant
- W A and B have to be singular
- W $ABx = 0$ has more than one solution (Lösungsschar)
- C $ABx = c$ can have no, one and ∞ many solutions, if $c \in \mathbb{R}^2, c \neq 0$
- W It has to be $A = 0$ or $B = 0$

- Which of the following statements are correct for an arbitrary $n \times n$ -matrix A and for arbitrary n ?
- W $\det(2A) = 2\det(A)$
- W $\det(-A) = \det(A)$
- C $\det(A^4) = \det(A)^4$
- W Let A be a triangular matrix with the property $a_{i,j} = 0$ for $i + j > n + 1$ (so there are zeros at the bottom right). The determinant can be calculated using the formula $\det(A) = a_{1,1} \cdot a_{2,2} \cdot \dots \cdot a_{n,1}$.

- Let $A, P, Q \in \mathbb{R}^{n \times n}$ where P is permutation matrix and Q is a unitary matrix.
- W $\det(PA) = \det(A)$
- C $\det(PAP) = \det(A)$
- W $\det(QA) = \det(A)$

Eigenvalue/Eigenvectors

- C with $\mathcal{X}_A(\lambda) = (\lambda - 1)^3 + 3$ is $A \in \mathbb{E}^{3 \times 3}$ invertible 0 isn't eigenvalue $\Rightarrow A$ is regular
- W Let v_1 and v_2 be eigenvectors of A , so is $v_1 + v_2$ an eigenvector
- C If $A \in \mathbb{E}^{n \times n}$ and $\mathcal{X}_A(\lambda) = (\lambda - 1)^n + 2$, A is invertible
- C Similar matrices have the same eigenvalues
- W Similar matrices have the same eigenvectors
- W Every $n \times n$ matrix has linear independent eigenvectors
- W Eigenvectors, which correspond to the same eigenvalue are always linear dependant
- C If a real matrix has a eigenvector, it follows that the matrix has infinity many eigenvectors
- C Every rotation in $\mathbb{R}^3$ has the eigenvalue $\lambda = 1$
- W If λ_1 with v and λ_2 with w , so is $(\lambda_1 + \lambda_2)$ a eigenvalue with eigenvector $v + w$
- Let $A \in \mathbb{R}^{3 \times 3}$ with eigenvalues $\lambda_1, \lambda_2, \lambda_3$
- C A is diagonalisable if all eigenvalues are different
- W If A is diagonalisable, all eigenvalues have to be different
- W A is diagonalisable if it has 3 eigenvectors They have to be linearly independent
- C If $\lambda_1 = 2, \lambda_2 = -2, \lambda_3 = 1$ and $B = A^3 - 3A^3$ then is B diagonalisable
- W If $AP = PD$ and D is a diagonal matrix then the columns of P are eigenvectors of A Only if the eigenvalues of A are on the diagonal of D

- Let $A \in \mathbb{R}^{n \times n}$ be positiv definite and symmetric. Further, let $\lambda_1, \dots, \lambda_n$ be the eigenvalues to the eigenvectors $v_1, \dots, v_n$
- W A^2 has at least one eigenvalue with a strictly positive imaginary part
- C it holds that $\lambda_j > 0$ for all $j = 1, \dots, n$
- W A has at least one eigenvalue which satisfies: $\text{geom. mult.} < \text{alg. mult.}$
- W The eigenvalues are pairwise distinct: $\lambda_j \neq \lambda_i$, if $j \neq i$
- C There exist positiv real numbers $\alpha > 0$ such that $v^T A v \geq v^T v$ for all $v \in \mathbb{R}^n$

- Let $A \in \mathbb{E}^{2 \times 2}$ with $\text{Rank}(A) = 1$ and $\text{Trace}(A) = 5$. What are the eigenvalues
- W 1 is an eigenvalue
- C 0 is an eigenvalue
- W 2 is an eigenvalue
- W -5 is an eigenvalue
- C 5 is an eigenvalue Since $\det(A) = 0$ and $\text{trace}(A) = 5 \Rightarrow 0 = \lambda_1 \cdot \lambda_2$ and $5 = \lambda_1 + \lambda_2$

Decompositions

- C A matrix $A \in \mathbb{R}^{n \times n}$ with n eigenvalues has 2^n normed spectral decompositions since the sign can be changed n -times
- Let $A \in \mathbb{R}^{m \times n}$ $m \geq n$ be a matrix with rank k . Denote the QR-decomposition of A as $A = QR$, where $Q \in \mathbb{R}^{m \times k}$ has orthonormal columns, and $R \in \mathbb{R}^{k \times n}$ is an upper (right) triangular matrix. Which one of the following statements is always true?
- W $\text{Rank } A < \text{Rank } R$
- W $QQ^T = I$
- W If A has linearly independant columns, we have $\text{Rank } R = m$
- C If A has linearly independant columns, we have $\text{Rank } R = n$
- Let $A \in \mathbb{R}^{m \times n}$ with linearly independant columns and $A = Q_1 R_1 = Q_2 R_2$, two QR-Decompositions of A
- C $Q_1^T Q_2$ is orthogonal
- C $Q_1^T Q_2$ is a upper and lower triangular matrix and therefore a diagonal matrix
- W $Q_1^T Q_2 = I$
- W $\text{Rank}(R_1) = m$
- C $\text{Rank}(R_1) = n$
- W $\text{Rank}(R_2) = m$
- C $\text{Rank}(R_2) = n$
- C $Q_1^T Q_2$ is regular
- Let $A, B \in \mathbb{R}^{n \times m}$, B is regular and $B = QR$
- W $f: \mathbb{R}^n \mapsto \mathbb{R}, x \mapsto \|Ax\|_2$, is a norm in $\mathbb{R}^n$
- C If βk , such that A^k is invertible, so is A not invertible
- C $\forall x \in \mathbb{R}^n, \|Ax\|_2 \leq \|A\|_2 \cdot \|x\|_2$
- W $\det(B) = \det(R)$
- C $\|B\|_2 = \|R\|_2$
- C $g: \mathbb{R}^n \mapsto \mathbb{R}, x \mapsto \|Qx\|_2$, is a norm in $\mathbb{R}^n$
- W AB is regular, but BA is not necessarily
- W $\|AB\|_2 \leq \|A\|_2$

Basis

- W The transformation matrix of a basis transformation between orthonormal bases is the identity matrix. The transformation matrix of base transformation between orthonormal bases is orthogonal, the identity matrix is only one possibility.
- C The inverse of the transformation matrix of a base transformation between orthonormal bases is its Hermitian transpose.
- C $A \in \mathbb{R}^{n \times n}$ is an orthogonal matrix if and only if its columns form an orthonormal basis of $\mathbb{R}^n$ with respect to the Euclidean scalar product.
- C The change of basis matrix is unitary (if $\mathbb{E} = \mathbb{C}$) or orthonormal (if $\mathbb{E} = \mathbb{R}$) if both bases are orthonormal.

Procedures

- W The Gram-Schmidt orthogonalization method can be used to compute an equally large set of linearly independent vectors from a set of linearly dependent vectors.
- W Let $v_1, \dots, v_n \subset \mathbb{R}^n$ be a set of n vectors. Using the Gram-Schmidt process, we can always produce n unit-length and pairwise orthogonal vectors. Gram Schmidts needs linear independent vectors

- Let $A \in \mathbb{R}^{n \times n}$, $m < n$. Let $Ax = b$ be a system of linear equations and let x be a solution in the least squares sense. Which statement is always correct?
- W The vector (bAx) is orthogonal to the row space of A .
- C The vector (bAx) is orthogonal to the column space of A . $\Rightarrow$ normal equations
- W z is in the null space of A .
- W The solution x does not always exist

Kernel/Image

- W If the nullspace of an 8×7 matrix is 5-dimensional, the rowspace has dimension 3 $n - \dim(\text{Ker}(A)) = 7 - 5 = 2 = \dim(\text{Im}(A))$

- Let $A \in \mathbb{R}^{m \times n}$ be such that $Ax = 0$ has only the trivial solution. Then it holds that:
- C $\dim \text{Im}(A) = n$
- W $\dim \text{Im}(A) = 1$
- C $\dim \text{Ker}(A) = 0$
- W $\dim \text{Ker}(A) = 1$ The kernel of A is exactly the solution set of the system of equations $Ax = 0$. Since $Ax = 0$ has only the trivial solution, $\dim \text{Ker}(A) = 0$. Furthermore, it holds that $\dim \text{Ker}(A) + \dim \text{Im}(A) = n$. Therefore $\dim \text{Im}(A) = n$.

- Which of the following statements with $A \in \mathbb{R}^{n \times n}$ is generally true
- C $\text{im}(A) = \text{im}(2A)$
- C $\ker(A) \cap \ker(2A)$
- W $\text{im}(A) = \ker(A^2)$
- W $\text{im}(A) = \text{im}(A + I)$
- W $\text{im}(A) = \text{im}(A^T)$
- W $\ker(A) \cap \ker(A^2)$
- W $\ker(A) \cap \ker(A + I)$
- W $\ker(A) \cap \ker(A^T)$

Proofs

- 1) Prove that $A^H A$ and $A A^H$ have the same eigenvalues
- We have that $A A^H$ and $A A^H$ are similar with $T = A$ and $T^{-1} = A^H$
Therefore by Satz 9.7 they have the same eigenvalues (and also the same trace and det)
- 2) Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. Prove that, if n is odd, that at least one of the matrices $(Q + I)$ and $(Q - I)$ singular.
- Let $K_Q(x)$ be the characteristic polynomial of Q . λ is an eigenvalue of $Q \Leftrightarrow \lambda$ is a root of $K_Q(x)$
 Q is orthogonal $\Rightarrow |\lambda| = 1$
By Lemma 9.2 we have: λ is eigenvalue $\Rightarrow (A - \lambda I)$ is singular
therefore with $\lambda = \pm 1$ at least one of them has to be singular
- 3) Prove that for an orthogonal matrix Q it holds that $\|Qx\|_2 = \|x\|_2$
- $\|Qx\|_2 \stackrel{1}{=} \sqrt{\langle Qx, Qx \rangle} \stackrel{2}{=} \sqrt{\langle Qx \rangle^T Qx} \stackrel{S.2.6}{=} \sqrt{x^T Q^T Qx} \stackrel{S.2.20}{=} \sqrt{x^T I x} \stackrel{2}{=} \sqrt{\langle x, x \rangle} \stackrel{1}{=} \|x\|_2$
1 = def of norm, 2 def of scalar product
- 4) Prove that, if λ is an eigenvalue of orthogonal Q then $\lambda = \pm 1$
- $Qv = \lambda v(1) \Leftrightarrow \|Qv\| = \|\lambda v\| \stackrel{2}{\Leftrightarrow} \|v\| = \|\lambda v\| \stackrel{N2}{\Leftrightarrow} \|v\| = |\lambda| \|v\| \Leftrightarrow |\lambda| = 1$
1 = def eigenvalue, 2 = as proven before, N2 = norm is homogeneous
- 5) Prove that for a arbitrary matrix A with its eigenvalue λ it holds that $(A - \lambda I)$ is singular
- Let v be the eigenvector to the corresponding λ in $(A - \lambda I)$
- $(A - \lambda I)v = (Av - \lambda Iv) = Av - \lambda v \stackrel{1}{=} \lambda v - \lambda v = 0v$
As v is eigenvector we have
- $v \neq 0(2) \Rightarrow \lambda = 0 \stackrel{L.9.6}{\Leftrightarrow} (A - \lambda I)$ is singular
1 = as v is eigenvalue, 2 = def eigenvalue
- 6) Let $A \in \mathbb{R}^{n \times n}$ be a real matrix and $x \in \mathbb{R}^n$ be a vector. Prove that: $\|Ax\|_2 \geq \sigma_{\min} \|x\|_2$. Where $\sigma_{\min}$ is the smallest singular value of A
- Let $A = U \Sigma V^T$ be the SVD of A
- $\|Ax\|_2 = \|U \Sigma V^T x\|_2 \stackrel{1}{=} \|\Sigma V^T x\|_2 \stackrel{2}{=} \|\Sigma_{\min} V^T x\|_2 = \|\Sigma_{\min} I V^T x\|_2 \stackrel{N2}{=} \|\Sigma_{\min} V^T x\|_2 = \|V^T x\|_2 \stackrel{3}{=} |\sigma_{\min}| \|x\|_2$
1 = U is orthogonal and proof 3, 2 = $\Sigma_{\min} = \text{diag}(\sigma_{\min}, \dots)$, 3 = V^T is orthogonal and proof 3
- 7) Prove that for a regular matrix A the two SLE $Ax = b$ and $A^T Ax = A^T b$ yield the same solution
- $A^T Ax = A^T b \stackrel{1}{\Leftrightarrow} A^{-T} A^T Ax = A^{-T} A^T b \Rightarrow Ax = b$
1 = Since A is regular
- $\det(A) \neq 0 \stackrel{S.8.9}{\Rightarrow} \det(A^T) \neq 0$, hence A^T is regular as well
- 8) For $A \in \mathbb{R}^{n \times n}$ holds that $A = A^T$. Prove that all eigenvalues of A^{2k} $k \in \mathbb{N}$ are not negative.
- Lets define the SVD as follows: $A = V \Sigma V^{-1}$
Further we know A is orthogonal
- $A = V \Sigma V^{-1} \stackrel{S.2.20}{=} A = V \Sigma V^T$
One has to prove inductively that $A^n = V \Sigma^n V^T$
- B.C for $n = 1$ it holds that: $A = V \Sigma^1 V^T$
- I.H Assume it holds for any $k \in \mathbb{N}$

I.S $k \mapsto k+1$: $A^{k+1} = AA^k \stackrel{\text{I.H}}{=} AV\Sigma^k VT \stackrel{n=1}{=} V\Sigma VT V\Sigma^k VT \stackrel{\text{S.2.20}}{=} V\Sigma\Sigma^k VT = V\Sigma^{k+1} VT$
The eigenvalue for A^n can be found in the diagonal of Σ^n . With σ_q as the original eigenvalues of A , one can write:
 $\Sigma^{2k} = \text{diag}(\sigma_1^{2k} \dots \sigma_n^{2k})$ **Therefore no eigenvalue can be negative.**
9) Prove that $A^H Ax = A^H b$ has infinitely many solutions.
 $A^H Ax = A^H b \Rightarrow A^H (Ax - b) = 0 \Rightarrow$
 $A^H (Ax - (b_\perp + b_\parallel)) = A^H ((Ax - b_\perp) - b_\parallel) \stackrel{1}{=}$
 $A^H (-b_\perp) \Rightarrow -(A^H b_\perp) \stackrel{2}{=} 0$
1 as $b_\perp \in \mathcal{R}(A)$ it follows: $Ax - b_\perp = 0$
2 $b \in \mathcal{N}(A^H)$

Therefore at least one solution exists. Since $\text{rank}(A) < n \Leftrightarrow \dim(\mathcal{N}(A)) > 0$ and as every solution of the system is in $S_p + \alpha S_h$ for any α . Where S_p is a arbitrary particular solution and S_h is the homogeneous solution.
9) Prove Satz 9.7. Namely, proof for any two similar matrices that the characteristict polynomial and det is equal
Assume A and C are similar. Therefore we have that $C = T^{-1}AT$.
Characteristic Polynomial: $\chi_C(\lambda) \stackrel{1}{=} (C - \lambda I) \stackrel{2}{=}$
 $\det(T^{-1}(AT - \lambda T)) \stackrel{\text{S.8.7}}{=} \det(T^{-1})\det(AT - \lambda T) =$
 $\det(AT - \lambda T)\det(T^{-1}) \stackrel{\text{S.8.7}}{=} \det((AT - \lambda T)T^{-1}) =$
 $\det(ATT^{-1} - \lambda TT^{-1}) = \det(A - \lambda I) \stackrel{1}{=} \chi_A(\lambda)$
1 = def of characteristic polynomial

2 = since $C = T^{-1}AT$
Det:
 $\det(C) \stackrel{1}{=} \det(T^{-1}AT) \stackrel{\text{S.8.7}}{=} \det(T^{-1}) \cdot \det(A) \cdot \det(T) =$
 $\det(T^{-1}) \cdot \det(T) \cdot \det(A) \stackrel{\text{C.8.8}}{=} 1 \cdot \det(A) = \det(A)$
1 = def of $C = T^{-1}AT$
10) let V and W be two vectors spaces. Let $\phi : V \mapsto W$ be a linear mapping. Show, that $\text{Im}(\phi)$ is a subspace of W
1 $\text{im}(\phi)$ isn't empty: $0 = \phi(0)$ since ϕ is linear
2 For $x, y \in \text{Im}(\phi)$ holds that $x + y \in \text{im}(\phi)$: $\exists a, \exists b$ such that $\phi(a) = x$ and $\phi(b) = y$ Therefore, $x + y = \phi(a) + \phi(b) = \phi(a + b) \in \text{Im}(\phi)$ since ϕ is linear
3 For $x \in \text{Im}()$ and $\alpha \in \mathbb{R}$ holds that $\alpha x \in \text{Im}()$:

$\exists a$, **such that $\phi(a) = x$ Therefore,**
 $\alpha x = \alpha \phi(a) = \phi(\alpha a) \in \text{Im}(\phi)$ **since ϕ is linear**
11) Let $B = (1, x, x^2)$ and $B' = (x + 1, x - 1, x^2)$.The columns (spalten) of T are the elements of B' in the basis B Then $T_{B' \rightarrow B} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ by inverting
 $T_{B \rightarrow B'}$ **we get $T_{B' \rightarrow B} = \begin{pmatrix} 0.5 & 0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ The**
mapping matrix D' is then given by
 $D' = T_{B \rightarrow B'}DT_{B' \rightarrow B} = \begin{pmatrix} 0.5 & 0.5 & 1 \\ -0.5 & -0.5 & 1 \\ 0 & 0 & 0 \end{pmatrix}$